Endoscopic dynamic scene reconstruction: comparison with the state of the art

Third-party numbers are copied from each cited paper (arXiv HTML); a blank cell means not reported. PSNR/SSIM/LPIPS on held-out frames; SSIM printed in % by some papers is converted to 0–1.
AbsRel: ours is the rendered depth vs. SGBM stereo depth after per-frame median scaling; SurgicalGS's is vs. RAFT stereo depth (different definition). FPS, time and memory are on each paper's own GPU.

Method	Venue	Depth input	PSNR↑	SSIM↑	LPIPS↓	AbsRel↓	FPS	Train	Mem.	GPU	Source	Note
A. StereoMIS P1, frames ∼800–1000 (same clip as this work). Different papers ran different baselines on this clip; rows are grouped by source.												
EndoNeRF	MICCAI'22		21.49	0.622	0.360		0.2	5 h	4 GB	RTX 4090	Endo-4DGS [v3/v4] Tab. 2	re-run by Endo-4DGS authors
EndoSurf	MICCAI'23		29.87	0.809	0.303		0.04	8 h	14 GB	RTX 4090	Endo-4DGS [v3/v4] Tab. 2	re-run by Endo-4DGS authors
LerPlane-32k	MICCAI'23		30.80	0.826	0.174		1.7	7 min	19 GB	RTX 4090	Endo-4DGS [v3/v4] Tab. 2	re-run by Endo-4DGS authors
Endo-4DGS	MICCAI'24	mono (DA-S)	32.69	0.850	0.148		100	7 min	4 GB	RTX 4090	Endo-4DGS [v3/v4] Tab. 2	v1/v2 of the paper report 31.46 / 0.829 / 0.175
EndoNeRF	MICCAI'22		31.12	0.846	0.148					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
EndoSurf	MICCAI'23		31.29	0.865	0.168					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
ForPlane-32k	MedIA		32.19	0.843	0.143					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
Endo-4DGS	MICCAI'24	mono (DA)	32.17	0.850	0.192					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
EndoGaussian	arXiv'24		32.79	0.858	0.182					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
Deform3DGS	MICCAI'24		33.35	0.871	0.169					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
SurgicalGS	arXiv'24		32.98	0.867	0.172					A100	Endo-PairGS Tab. 2	P1 [802,1001]; all retrained
Endo-PairGS	IJCARS'26	Splatt3R pairs	33.56	0.871	0.146		100	∼10 min	4 GB	A100	Endo-PairGS Tab. 2	values read via page summary; equal to Endo-TSR Tab. 1
Endo-TSR	arXiv'26	stereo (Foundation-Stereo)	38.45	0.947	0.074						Endo-TSR Tab. 1	uses stereo depth; baselines copied from Endo-PairGS
EndoNeRF	MICCAI'22		30.500	0.880	0.078		0.2			RTX 4090	Endo-G2T Tab. 1	re-run status not stated
EndoSurf	MICCAI'23		31.221	0.890	0.071		0.04			RTX 4090	Endo-G2T Tab. 1	re-run status not stated
Endo-4DGS	MICCAI'24	mono (DA)	32.188	0.898	0.066		100			RTX 4090	Endo-G2T Tab. 1	re-run status not stated
ST-Endo4DGS	arXiv'24		32.900	0.905	0.060		123			RTX 4090	Endo-G2T Tab. 1	
Endo-G2T	arXiv'25		33.580	0.914	0.056		148			RTX 4090	Endo-G2T Tab. 1	FPS raster-only
SurgicalGS	arXiv'24		31.60	0.854	0.263		183			RTX 4090	EndoWave Tab. II	other rows copied from Endo-4DGS
EndoWave	arXiv'25		33.26	0.9126	0.075		77			RTX 4090	EndoWave Tab. II	text gives LPIPS 0.0554
Endo-4DGS, reproduced	–	mono (DA-B)	31.98±0.09	0.851±0.001	0.178±0.003	0.234±0.004	38	16.0 min	0.6 GB*	RTX 4060 Ti	this work (3 seeds)	P1 801-1000; robot poses
+ corrected FoV (DA)	–	mono (DA-B)	32.33±0.14	0.854±0.002	0.186±0.002	0.235±0.039	35	17.0 min	0.6 GB*	RTX 4060 Ti	this work (3 seeds)	
+ corrected FoV, DA init only	–	mono (DA-B)	32.62±0.13	0.860±0.002	0.165±0.001	0.376±0.048		15.6 min		RTX 4060 Ti	this work (3 seeds)	best PSNR, worst geometry
+ corrected FoV, MoGe-2 + depth loss	–	mono (MoGe-2)	31.87±0.13	0.844±0.002	0.216±0.004	0.155±0.003		17.1 min		RTX 4060 Ti	this work (3 seeds)	
Ours: corrected FoV + MoGe-2 init	**–**	**mono (MoGe-2)**	**32.40±0.10**	**0.853±0.001**	**0.190±0.001**	**0.148±0.001**	**38**	**15.9 min**	**0.6 GB***	**RTX 4060 Ti**	**this work (3 seeds)**	***render memory**
B. StereoMIS, other sequences and protocols (not comparable with Part A or with each other; clip shown per row or in the note).												
EndoNeRF	MICCAI'22		28.86	0.742	0.27		0.03	∼6 h		RTX A5000	Deform3DGS Tab. 1	3 clips P2_7+P3; SSIM printed in %
LerPlane	MICCAI'23		29.46	0.777	0.20		1.52	∼10 min		RTX A5000	Deform3DGS Tab. 1	3 clips P2_7+P3
EndoGaussian	arXiv'24		30.25	0.828	0.21		134.5	151 s		RTX A5000	Deform3DGS Tab. 1	3 clips P2_7+P3
Deform3DGS	MICCAI'24		30.48	0.827	0.21		330.4	66 s		RTX A5000	Deform3DGS Tab. 1	3 clips P2_7+P3
EndoGaussian	arXiv'24		30.39	0.838	0.21					RTX 3060	ColorGS Tab. 1	Deform3DGS protocol
Deform3DGS	MICCAI'24		30.68	0.847	0.23					RTX 3060	ColorGS Tab. 1	Deform3DGS protocol
ColorGS	arXiv'25		32.64	0.896	0.14					RTX 3060	ColorGS Tab. 1	Deform3DGS protocol
EndoNeRF (P2_7 / P3)	MICCAI'22		28.87 / 28.33	0.802 / 0.801	0.259 / 0.287		0.06			V100	Diff2DGS Tab. II	RMSE 4.354 / 4.165
LerPlane (P2_7 / P3)	MICCAI'23		29.53 / 29.41	0.784 / 0.781	0.205 / 0.213		0.97			V100	Diff2DGS Tab. II	RMSE 8.356 / 7.983
EndoGaussian (P2_7 / P3)	arXiv'24		31.11 / 31.03	0.873 / 0.871	0.127 / 0.136		90.3			V100	Diff2DGS Tab. II	RMSE 9.793 / 8.932
Deform3DGS (P2_7 / P3)	MICCAI'24		31.83 / 30.91	0.878 / 0.877	0.129 / 0.131		214.4			V100	Diff2DGS Tab. II	RMSE 10.036 / 8.482
SurgicalGS (P2_7 / P3)	arXiv'24		31.63 / 31.49	0.883 / 0.878	0.152 / 0.165		154.8			V100	Diff2DGS Tab. II	RMSE 2.747 / 2.536
Diff2DGS (P2_7 / P3)	arXiv'26		34.72 / 34.08	0.912 / 0.903	0.113 / 0.131		212.3			V100	Diff2DGS Tab. II	RMSE 3.874 / 3.717
EndoNeRF (intestine / liver)	MICCAI'22		28.694 / 27.738	0.783 / 0.712	0.279 / 0.345		0.06		13 GB	RTX 3090	SurgicalGaussian Tab. 1	2 fast-motion clips; tool pixels excluded
EndoSurf (intestine / liver)	MICCAI'23		29.660 / 28.941	0.853 / 0.820	0.204 / 0.248		0.08		14 GB	RTX 3090	SurgicalGaussian Tab. 1	
LerPlane (intestine / liver)	MICCAI'23		29.441 / 28.852	0.822 / 0.793	0.206 / 0.254		1.45		19 GB	RTX 3090	SurgicalGaussian Tab. 1	
EndoGaussian (intestine / liver)	arXiv'24		29.024 / 26.174	0.805 / 0.728	0.213 / 0.295		200		2 GB	RTX 3090	SurgicalGaussian Tab. 1	
SurgicalGaussian (intestine / liver)	arXiv'24	stereo (STTR-light)	31.496 / 31.668	0.890 / 0.893	0.145 / 0.135		140		3 GB	RTX 3090	SurgicalGaussian Tab. 1	
EndoNeRF (intestine / liver)	MICCAI'22		30.510 / 27.370	0.833 / 0.680	0.153 / 0.316					RTX A6000	SGS Tab. 1	P2_7 + P2_8 segments
Deform3DGS (intestine / liver)	MICCAI'24		32.589 / 32.929	0.897 / 0.900	0.131 / 0.110					RTX A6000	SGS Tab. 1	
EndoGS (intestine / liver)	arXiv'24		35.239 / 34.540	0.934 / 0.932	0.060 / 0.051					RTX A6000	SGS Tab. 1	
SGS (intestine / liver)	WACV'26	mono (Omnidata)	34.669 / 34.390	0.936 / 0.934	0.048 / 0.047		423		2 GB	RTX A6000	SGS Tab. 1	
EndoNeRF	MICCAI'22	stereo (RAFT)	28.79	0.809	0.266	0.0315	0.06			RTX 3090	SurgicalGS Tab. I	sequences not stated; AbsRel vs RAFT depth
EndoSurf	MICCAI'23	stereo (RAFT)	29.36	0.861	0.211	0.0189	0.05			RTX 3090	SurgicalGS Tab. I	
LerPlane	MICCAI'23	stereo (RAFT)	29.09	0.789	0.179	0.0354	0.95			RTX 3090	SurgicalGS Tab. I	
EndoGaussian	arXiv'24	stereo (RAFT)	31.02	0.878	0.132	0.0292	130.2			RTX 3090	SurgicalGS Tab. I	

Method	Venue	Depth input	PSNR↑	SSIM↑	LPIPS↓	AbsRel↓	FPS	Train	Mem.	GPU	Source	Note
Deform3DGS	MICCAI'24	stereo (RAFT)	31.61	0.888	0.135	0.0330	308.7			RTX 3090	SurgicalGS Tab. I	
SurgicalGS	arXiv'24	stereo (RAFT)	31.54	0.885	0.148	0.0082	214.1			RTX 3090	SurgicalGS Tab. I	
ForPlane (P1_1 / P1_2)	MedIA		28.47 / 28.95	0.824 / 0.829	0.258 / 0.239		1.2	215 s		RTX 4090	EH-SurGS Tab. I	184 / 70 images; all 3000 it., 3 runs
Deform3DGS (P1_1 / P1_2)	MICCAI'24		34.54 / 34.87	0.896 / 0.911	0.170 / 0.156		357.7	92 s		RTX 4090	EH-SurGS Tab. I	
Endo-4DGS (P1_1 / P1_2)	MICCAI'24	mono (DA)	33.52 / 34.18	0.884 / 0.904	0.180 / 0.150		95.7	240 s		RTX 4090	EH-SurGS Tab. I	
LGS (P1_1 / P1_2)	–		23.46 / 25.47	0.842 / 0.819	0.270 / 0.315		142.6	144 s		RTX 4090	EH-SurGS Tab. I	
SurgicalGaussian (P1_1 / P1_2)	arXiv'24		28.91 / 31.26	0.831 / 0.859	0.366 / 0.251		304.0	159 s		RTX 4090	EH-SurGS Tab. I	
EH-SurGS (P1_1 / P1_2)	arXiv'25		34.73 / 35.09	0.898 / 0.913	0.177 / 0.155		380.3	111 s		RTX 4090	EH-SurGS Tab. I	
Deform3DGS (P1_1 / P1_2)	MICCAI'24		28.55 / 32.28	0.846 / 0.887						RTX 4090	Endo-4DGX Tab. 2	P1 16400-16697 / 7379-7499; uneven illumination
Endo-4DGS (P1_1 / P1_2)	MICCAI'24	mono (DA)	26.98 / 32.48	0.817 / 0.890						RTX 4090	Endo-4DGX Tab. 2	
EndoGaussian (P1_1 / P1_2)	arXiv'24		27.61 / 32.20	0.841 / 0.896						RTX 4090	Endo-4DGX Tab. 2	
Endo-4DGX (P1_1 / P1_2)	MICCAI'25		28.73 / 32.70	0.853 / 0.899			61		6.5 GB	RTX 4090	Endo-4DGX Tab. 2	
EndoNeRF (P2-2 / P2-5 / P3)	MICCAI'22		28.91 / 26.49 / 29.85	0.703 / 0.688 / 0.751	0.261 / 0.274 / 0.209					A100	Endo-PairGS Tab. 2	all retrained
EndoSurf (P2-2 / P2-5 / P3)	MICCAI'23		30.01 / 26.51 / 30.46	0.780 / 0.738 / 0.811	0.198 / 0.277 / 0.188					A100	Endo-PairGS Tab. 2	
ForPlane-32k (P2-2 / P2-5 / P3)	MedIA		28.01 / 28.48 / 31.10	0.659 / 0.759 / 0.801	0.286 / 0.170 / 0.139					A100	Endo-PairGS Tab. 2	
Endo-4DGS (P2-2 / P2-5 / P3)	MICCAI'24	mono (DA)	31.45 / 29.06 / 31.21	0.812 / 0.817 / 0.831	0.141 / 0.152 / 0.155					A100	Endo-PairGS Tab. 2	
EndoGaussian (P2-2 / P2-5 / P3)	arXiv'24		29.82 / 28.28 / 31.26	0.759 / 0.782 / 0.821	0.169 / 0.173 / 0.166					A100	Endo-PairGS Tab. 2	
Deform3DGS (P2-2 / P2-5 / P3)	MICCAI'24		30.69 / 29.16 / 31.70	0.789 / 0.813 / 0.834	0.144 / 0.147 / 0.160					A100	Endo-PairGS Tab. 2	
SurgicalGS (P2-2 / P2-5 / P3)	arXiv'24		30.19 / 29.07 / 31.97	0.775 / 0.805 / 0.841	0.178 / 0.168 / 0.155					A100	Endo-PairGS Tab. 2	
Endo-PairGS (P2-2 / P2-5 / P3)	IJCARS'26	Splatt3R pairs	32.16 / 29.75 / 32.47	0.833 / 0.846 / 0.871	0.100 / 0.103 / 0.110		100		4 GB	A100	Endo-PairGS Tab. 2	
Endo-TSR (P2-2 / P2-5 / P3)	arXiv'26	stereo (Foundation-Stereo)	34.83 / 31.99 / 36.19	0.879 / 0.894 / 0.939	0.065 / 0.067 / 0.045						Endo-TSR Tab. 1	3 seeds
EndoNeRF	MICCAI'22		31.922	0.857	0.146					RTX 4090	Endo-GSMT Tab. 1	5 segments; SLAM poses; flow-aligned metrics
LerPlane-32k	MICCAI'23		31.679	0.845	0.113					RTX 4090	Endo-GSMT Tab. 1	
EndoGS	arXiv'24		32.819	0.907	0.099					RTX 4090	Endo-GSMT Tab. 1	
EndoGaussian	arXiv'24		29.191	0.827	0.181					RTX 4090	Endo-GSMT Tab. 1	
Endo-4DGS	MICCAI'24	mono (DA)	31.580	0.862	0.124					RTX 4090	Endo-GSMT Tab. 1	
Deform3DGS	MICCAI'24		32.209	0.863	0.124					RTX 4090	Endo-GSMT Tab. 1	
Endo-GSMT	MICCAI'25		34.703	0.917	0.060					RTX 4090	Endo-GSMT Tab. 1	
C. EndoNeRF dataset (cutting / pulling).												
EndoNeRF	MICCAI'22		35.84 / 35.43	0.942 / 0.939	0.057 / 0.064		0.2	6 h	4 GB	RTX 4090	Endo-4DGS Tab. 1	
EndoSurf	MICCAI'23		34.89 / 34.91	0.952 / 0.955	0.107 / 0.120		0.04	7 h	17 GB	RTX 4090	Endo-4DGS Tab. 1	
LerPlane-32k	MICCAI'23		34.66 / 31.77	0.923 / 0.910	0.071 / 0.071		1.5	8 min	20 GB	RTX 4090	Endo-4DGS Tab. 1	
Endo-4DGS	MICCAI'24	mono (DA-S)	36.56 / 37.85	0.955 / 0.959	0.032 / 0.043		100	4 min	4 GB	RTX 4090	Endo-4DGS Tab. 1	v1/v2: 36.84 / 37.08
EndoSurf	MICCAI'23		34.981 / 35.004	0.953 / 0.956	0.106 / 0.120			9 h		RTX 3090	EndoSurf Tab. 1	RMSE 0.835 / 1.165 mm
EndoNeRF	MICCAI'22		34.186 / 34.217	0.932 / 0.938	0.151 / 0.160		0.04		15 GB	RTX 3090	SurgicalGaussian Tab. 1	
LerPlane	MICCAI'23		35.580 / 36.241	0.955 / 0.950	0.101 / 0.102		1.02		20 GB	RTX 3090	SurgicalGaussian Tab. 1	
EndoGaussian	arXiv'24		38.287 / 37.308	0.962 / 0.958	0.058 / 0.070		190		2 GB	RTX 3090	SurgicalGaussian Tab. 1	
SurgicalGaussian	arXiv'24		37.505 / 38.783	0.961 / 0.970	0.062 / 0.049		80		4 GB	RTX 3090	SurgicalGaussian Tab. 1	
Endo-4DGS	MICCAI'24	mono (DA)	36.13 / 37.18	0.951 / 0.955	0.054 / 0.072					RTX 4090	EH-SurGS Tab. I	3000 it.
Deform3DGS	MICCAI'24		38.77 / 38.33	0.967 / 0.961	0.042 / 0.063					RTX 4090	EH-SurGS Tab. I	
EH-SurGS	arXiv'25		39.91 / 38.72	0.972 / 0.963	0.034 / 0.062		379.7 / 387.0	105 / 97 s		RTX 4090	EH-SurGS Tab. I	
Endo-4DGS	MICCAI'24	mono (DA)	36.56 / 37.85	0.955 / 0.959	0.032 / 0.043					A100	Endo-PairGS Tab. 3	identical to Endo-4DGS's own numbers
Endo-PairGS	IJCARS'26	Splatt3R pairs	38.33 / 38.96	0.967 / 0.967	0.020 / 0.022		100		4 GB	A100	Endo-PairGS Tab. 3	
Endo-TSR	arXiv'26	stereo (Foundation-Stereo)	40.96 / 39.34	0.980 / 0.967	0.019 / 0.035						Endo-TSR Tab. 1	
Endo-4DGS	MICCAI'24	mono (DA)	36.165 / 37.014	0.959 / 0.960	0.039 / 0.041		100			RTX 4090	Endo-G2T Tab. 1	
Endo-G2T	arXiv'25		40.080 / 38.290	0.982 / 0.970	0.007 / 0.016		148			RTX 4090	Endo-G2T Tab. 1	
EndoGS	arXiv'24		37.16 / 36.19	0.953 / 0.941	0.045 / 0.041					RTX 4090	EndoWave Tab. I	
SurgicalGS	arXiv'24		38.31 / 38.05	0.962 / 0.959	0.062 / 0.062		183			RTX 4090	EndoWave Tab. I	
EndoWave	arXiv'25		38.93 / 38.51	0.981 / 0.969	0.010 / 0.021		86			RTX 4090	EndoWave Tab. I	
HFGS	arXiv'24		37.83 / 38.44	0.969 / 0.968	0.022 / 0.043						HFGS Tab. 2	
This work	–										–	EndoNeRF data not available (request form)

Reading Part A. The same method varies by up to 0.7 dB between papers on this clip (Endo-4DGS: 31.98 ours, 32.17 Endo-PairGS, 32.19 Endo-G2T, 32.69 original), and its LPIPS ranges from 0.066 to 0.192, so differences below $\sim$1 dB across papers are not reliable. Within one protocol, our best model improves Endo-4DGS by +0.42 dB PSNR and cuts depth error by 37 %; the strongest monocular methods on this clip (Endo-PairGS 33.56, Endo-G2T 33.58, Deform3DGS 33.35 in Endo-PairGS, EndoWave 33.26) are about 1.2 dB higher in PSNR, but none report geometric accuracy. Endo-TSR (38.45) uses stereo depth and is not a monocular method.

Sources (arXiv unless noted): Endo-4DGS 2401.16416 (v3/v4); Endo-PairGS IJCARS 2026, doi:10.1007/s11548-026-03707-y; Endo-TSR 2609.32399; Endo-G2T 2511.21367; EndoWave 2510.23087; Deform3DGS 2405.17835; ColorGS 2508.18696; Diff2DGS 2602.18314; SurgicalGaussian 2407.05023; SGS 2503.04079; SurgicalGS 2410.09292; EH-SurGS 2501.01101; Endo-4DGX 2506.23308; Endo-GSMT MICCAI 2025 (paper 1655); EndoSurf 2307.11307; HFGS 2405.17872.